

The Two-Time Moving-Defect Problem for Dispersing Billiards

A go/no-go audit, dynamic graph currents, and a Fourier route

Working note

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Abstract

We isolate the estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of the scatterers. A tempting bridge from the Demers–Liverani projective cone to forward standard families fails: the moving-face coefficient uses a one-sided trace on a transversal terminal arc, whereas the cone controls stable-leaf integrals. We give a local counterexample to any inference of such a trace bound from the published cone seminorms. We prove a two-sided nuclear/recovery criterion which yields the desired product estimate, but emphasize that it is only a sufficient condition and is probably too strong for a graph delta. The intrinsic problem is instead two-parameter equidistribution of a centered graph current under $T^{-1} \times T$. A targeted literature audit points to an induced generalized-baker/Fourier-decay strategy: encode the collision SRB measure on a two-sided return scheme, prove Fourier decay for the induced slice by non-concentration or L^2 flattening, and lift the estimate through the exponential tower tail and the grazing blocks. We state the first falsifiable local estimate and the quantitative margins needed for this program. No unconditional CM2 theorem is claimed here.

1 The exact target

Let P_s be the Markov-normalized transfer operator on a fixed collision fiber, $P_s \mathbf{1} = \mathbf{1}$, let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_X \|f\|_{C^1}, \quad m, n \geq 0, \quad (1)$$

for some $\tau, \vartheta < 1$. The same estimate, uniformly for $s^{-1}(P_s - P_0)$, is required before a parameter limit can be exchanged with the two time sums.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (2)$$

Centering separate faces is neither invariant nor sufficient. Without (2), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \longmapsto w(h|_{I^+} - h|_{I^-}) \delta_I, \quad (3)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (3) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (4)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (5)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (4)–(5) and their transverse Hölder comparisons can bound $\|u|_I\|_{C^0}$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (4). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C \varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C \varepsilon |W|^{-q} \leq C \varepsilon^{1-q}$, proving (5). Replacing y by y' changes the integral by at most $C \varepsilon |y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (3) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C \chi^m \implies \|K_V P_0^m (f_+ - f_-)\|_Y \leq C' \chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (6)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (7)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (8)$$

for centered g and centered η . If K has the presentation (6), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (6), (7), and (8),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

Remark 3.3. *The theorem makes clear why a usual functional correlation estimate with error $O(\tau^m + \vartheta^n)$ is insufficient. CM2 requires a centered operator ideal estimate at the event time, not just mixing on either side. Wormell's specialized two-time conditional-mixing conjecture is one way to encode precisely this product structure.*

Remark 3.4 (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable–unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (6) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf_j \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j(W_j, \rho_j). \quad (9)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (8) once the constants in (9) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (10)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (6).

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (11)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (12)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (12) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;
- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (13)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (13), then it has a nuclear recovery presentation (6) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (13) therefore gives (6). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (13) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

6 The intrinsic dynamic graph-current formulation

Let $\gamma_b : I_b \rightarrow M$ parametrize one side of a complete physical face in block b , let $F_b \gamma_b$ denote its assembled terminal image, and let w_b be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m} \gamma_b(t)) f(T^n F_b \gamma_b(t)) w_b(t) dt. \quad (14)$$

Equivalently, if $J_{V,b}$ is the graph current on $M \times M$ induced by $t \mapsto (\gamma_b(t), F_b \gamma_b(t))$, then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (15)$$

Complete double centering removes the two marginals of $J_{V,b}$. The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under $T^{-1} \times T$ at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

Proposition 6.1 (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure ν_b on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant C^2 phase maps F ,*

$$|\widehat{F_* \nu_b}(\xi)| \leq C_b (1 + |\xi|)^{-\eta} \quad (16)$$

for some $\eta > 0$. Suppose also that, after excluding phase cancellation, the expanding frequencies generated by m past returns and n future returns are bounded below by

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

apart from cylinder pieces whose total centered mass is bounded by $C\tau_0^m\vartheta_0^n$. Then, for sufficiently regular centered cylinder observables,

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (17)$$

for some $\tau, \vartheta < 1$.

Proof. Decompose g and f into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale $c \max\{\Lambda_-^m, \Lambda_+^n\}$. Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(16) gives the product factor $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$. Summability of the Fourier coefficients of C^r tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless. $\square$

The proposition deliberately isolates the missing theorem. Uniform (16) cannot be inferred from one-sided mixing. It must be proved from non-concentration and nonlinearity of the induced inverse branches and the face graph. The L^2 -flattening framework and nonlinear-pushforward theorems of Baker–Khalil–Sahlsten and Baker–Banaji are plausible tools for this step. Wormell proves the corresponding Fourier mechanism for classes of generalized baker maps, but not the present two-time billiard statement.

6.1 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii R_j, R_i and centre distance $d > R_j + R_i$, put the source centre at the origin, the target centre at $(d, 0)$, and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let $r(\alpha) = |(d, 0) - q_j(\alpha)|$ and $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$. The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$.

Lemma 6.2 (The circular tangency face is genuinely curved). *At the collinear source point $\alpha = 0$, write $r_0 = d - R_j$. Then*

$$\varphi_\sigma''(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (18)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

Proof. Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r\sqrt{r^2 - R_i^2}},$$

equation (18) follows. On the tangency face, $|(d, 0) - q_j| \times v_\theta| = R_i$ while the longitudinal component has magnitude $\sqrt{r_0^2 - R_i^2}$; therefore $|\partial_\theta G| = 2R_i\sqrt{r_0^2 - R_i^2} > 0$. $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in (19) have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

6.2 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc's Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom-A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

Proposition 6.3 (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank R with*

$$\mu\{R > L\} \leq Ce^{-aL}.$$

Suppose a Fourier proof at frequency $|\xi|$ uses at most $N(\xi) \leq c_0 \log(2 + |\xi|)$ successive symbols. Assume that the subsystem with all ranks at most L satisfies, uniformly in the truncation,

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq Ce^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided $a, \eta > 0$ and the finite-truncation constants have the displayed exponential bound.

Proof. The mass of orbit words which see a rank larger than L is at most $CN(\xi)e^{-aL}$. On the complement apply the finite-truncation estimate. Choose $L = \delta \log |\xi|$ with $0 < \delta < \eta/c$ (with no upper restriction if $c = 0$). The good contribution is $O(|\xi|^{-\eta+c\delta})$ and the bad contribution is $O(\log |\xi| |\xi|^{-a\delta})$. Both have a positive power saving. $\square$

Thus the precise new input is a *uniform* singular version of Leclerc's quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails can then remove the truncation. What remains open is uniform control of the temporal-distance non-concentration constants as the truncation grows.

7 An induced QNL/Fourier program

The most economical current route is the following.

- (F1) Construct an exponential return scheme on a regular magnet and retain both one-sided codes. In local product coordinates the return map is a countable piecewise-smooth hyperbolic map with collision SRB Gibbs weights. An exact generalized-baker skew product is unnecessary; the appropriate template is the quantitative-nonlinearity/sum-product theorem for nonlinear Axiom-A maps.
- (F2) Lift each complete physical face to finitely or countably many smooth graph slices of the return rectangles. Artificial cuts cancel before the lift, and the two marginals are removed before estimating any branch.
- (F3) Prove a uniform nonlinearity/non-concentration statement for the phase family

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m}\gamma_b(t) + \zeta \cdot T^n F_b \gamma_b(t). \quad (19)$$

Either $|\Phi'|$ or $|\Phi''|$ must be large outside a set with a quantitative cylinder-mass bound. Exact affine resonance must be excluded; this may require a generic open class of tables rather than all finite-horizon tables.

- (F4) Apply nonlinear pushforward Fourier decay or L^2 flattening to the stable conditional Gibbs measure. Exponential return tails provide the moment bounds; billiard curvature must provide total nonlinearity or a uniform non-concentration substitute.
- (F5) Lift the return-time estimate to collision time. For the generic assembled trace bound, if block b has physical mass $M_b \lesssim 2^{-b}$ and Fourier constant C_b , the required grazing margin is

$$\sum_{b \geq 0} 2^{-b} C_b < \infty. \quad (20)$$

A polynomial deterioration $C_b \lesssim 2^{\kappa b}$ is acceptable only for $\kappa < 1$. Large grazing expansion can help the oscillatory estimate, but its distortion cost must be tracked in the same inequality. For the Markov-normalized first radial one-face graph, the fixed-atlas flux calculation is sharper: its grouped singular coefficient is $O(2^{-(3-\epsilon_*)b})$, while the weaker $O(2^{-(1-\epsilon_*)b})$ term is a regular two-dimensional interior/Jacobian source. If the oscillatory factorization preserves that separation, the graph-current margin improves to

$$\sum_{b \geq 0} 2^{-(3-\epsilon_*)b} C_b < \infty,$$

allowing $\kappa < 3 - \epsilon_*$. Establishing this preservation for all past words is part of the CM2 proof and cannot be assumed.

- (F6) Establish the same estimates uniformly for the assembled difference quotients

$$s^{-1}(P_s - P_0),$$

and then apply dominated convergence.

The recent symbolic construction of Climenhaga–Day gives a concrete finite-alphabet two-sided coding and product structure for every finite-horizon Sinai billiard map. It is useful as a geometric skeleton, but it constructs the measure of maximal entropy. CM2

concerns the collision SRB/flux measure, so its Gibbs weights, holonomies, and parameter-uniform distortion still have to be built into the induced scheme. This measure mismatch cannot be hidden.

A more faithful use of the smooth Fourier theory is a finite-horseshoe approximation. Truncate the return and homogeneity ranks at L , retain a fixed regular periodic word witnessing nonlinearity, and realize the resulting finite branch core as a smooth area-preserving Axiom-A basic set Ω_L . Put on Ω_L the equilibrium state for the induced unstable-Jacobian potential. Leclerc's theorem then applies at each fixed L . To pass to the billiard, however, one must prove uniformly

$$\mu\{\text{an orbit segment sees rank} > L\} \leq CNe^{-aL}, \quad (21)$$

$$(\mu_L \otimes \mu_L)\{|\Delta_L| \leq \sigma\} \leq Ce^{cL}\sigma^\Gamma, \quad (22)$$

and the analogous estimate for the weighted face slice. A nonzero periodic-point cocycle in a fixed central branch is a plausible uniform QNL witness, but the published smooth theorem does not state uniformity under this singular exhaustion. Equation (22), rather than the mere existence of a Young tower, is the principal new lemma.

The standard-pair equidistribution theorem of Bálint–Nándori–Szász–Tóth proves that one-dimensional singular initial measures can mix exponentially after a suitable smoothing argument. It supplies the future-time benchmark, not the product estimate: smoothing in the future-stable direction makes the past factor in (14) rougher, and reversing the smoothing exchanges the problem rather than solving it.

Finally, the large-essential-spectrum result of Butterley–Canestrari–Castorrini warns against replacing this dynamic program by a globally stronger Banach norm which controls every discontinuity trace. For piecewise two-dimensional systems such a norm can retain large essential spectrum. This is consistent with the local collar counterexample and favors an orbit-dependent oscillatory estimate.

7.1 Execution audit of the six links

The six links now have the following status.

- (1) *Return coding: available only as an input architecture.* Exponential Young structures and local product rectangles are standard for finite-horizon dispersing billiards. What has not been extracted in the literature is a parameter-uniform two-sided return model carrying the moving face and all its homogeneity labels with the constants needed below.
- (2) *SRB weights: conceptually closed, quantitatively unfinished.* Markov normalization identifies the collision invariant density and the induced Gibbs potential is the unstable-Jacobian potential. The recent finite-alphabet product construction for the measure of maximal entropy cannot replace these weights.
- (3) *Face lift: locally passed.* Artificial faces cancel before the coarea current is formed, the central radial tangency graph has nonzero coarea derivative, and Lemma 6.2 gives an explicit non-affinity certificate. A global bi-return partition of the face remains to be written.
- (4) *Quantitative nonlinearity: the unique principal gap.* Local curvature does not exclude cancellation after long balanced past/future words. The required statement is a billiard/countable-alphabet analogue of Leclerc's temporal-distance QNL, uniform over finite truncations.

- (5) *Fourier conversion: conditionally closed.* Proposition 6.1 gives CM2 once QNL yields uniform Fourier decay. The frequency scale is the maximum of the two hyperbolic scales, which still gives a product rate with half the Fourier exponent.
- (6) *Tower and grazing lift: a quantitative margin is identified.* Proposition 6.3 handles exponential return tails. The generic graph-current block margin is (20); the first radial one-face calculation potentially supplies the stronger exponent $3 - \epsilon_*$. Uniform difference-quotient QNL is still needed for parameter differentiation.

Accordingly the original six-step program does not yet prove CM2. It reduces the positive theorem to one new billiard-specific statement: a uniform temporal-distance non-concentration estimate for the double-centered radial face on finite return/homogeneity truncations, with constants growing slowly enough for Proposition 6.3 and the grazing margin.

7.2 A quantitative smoothing test

There is one falsifiable way to obtain (13). Replace a face delta in block b by a normalized collar of width ϵ . Suppose the regularized bridge and the collar error satisfy

$$\|K_{b,\epsilon}\|_{X_- \rightarrow Y_+} \leq CM_b \epsilon^{-p}, \quad (23)$$

$$|((K_{V,b} - K_{b,\epsilon})Q^m g)(\mathcal{U}_0^n f)| \leq CM_b \epsilon^\alpha \Lambda_-^m \Lambda_+^n \|g\| \|f\|_{C^1}. \quad (24)$$

If the regularized dynamics gives rates $\tau_0^m \vartheta_0^n$, choosing $\epsilon = e^{-a(m+n)}$ yields CM2 provided one can choose $a > 0$ so that

$$\tau_0 e^{ap} < 1, \quad \vartheta_0 e^{ap} < 1, \quad \Lambda_- e^{-a\alpha} < 1, \quad qquad \Lambda_+ e^{-a\alpha} < 1. \quad (25)$$

Equivalently, a genuine spectral/bunching margin is required. For billiards, unbounded grazing derivatives make (24) the difficult part. The current flux estimate $M_b \lesssim 2^{-b}$ and recovery bound $R_b \lesssim b$ control the sum over b , but do not supply the normal Hölder estimate in (24). Thus collar smoothing is a testable fallback, not a completed proof.

Proposition 7.1 (Uniform difference-quotient closure). *Suppose (11)–(12) hold uniformly with $K_{V,b}$ replaced by the assembled block of $s^{-1}(P_s - P_0)$, and each block converges weakly to $K_{V,b}$. Then the parameter difference quotient of the prescribed susceptibility is dominated by a summable $C\tau^m \vartheta^n$ majorant and converges term by term to the double series generated by K_V .*

Proof. Apply Theorem 3.2 uniformly in s and sum over b, m, n . Equation (12) and the two geometric series give an integrable majorant. Weak convergence on every block identifies the pointwise limit; dominated convergence completes the argument. $\square$

8 Next falsifiable tests

The first calculation should use one central physical face and a finite strip cutoff. After complete assembly and double centering, it should prove or disprove the local estimate

$$\left| \int_I g(T^{-m} \gamma(t)) f(T^n F \gamma(t)) w(t) dt \right| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}. \quad (26)$$

The proof attempt must analyze the phases (19). If a resonant branch admits covectors for which both the first and second phase derivatives remain small on positive slice mass,

the Fourier route fails for that table. If the non-concentration estimate holds uniformly, CM2 has a concrete proof route and the dyadic blocks should then be added using (20).

In parallel, one may still test the stronger mixed-cylinder condition (13). Success would imply CM2 immediately, but failure would only show that the graph current is non-nuclear; it would not refute (26).

The current status is therefore precise: future recovery is technically available, complete centering is algebraic, and the prescribed source is already independent and non-coboundary. The unresolved theorem is dynamic two-sided equidistribution of the complete terminal graph current. Existing projective cones, standard-family coupling, and anisotropic current spaces do not prove it. The best falsifiable route found in the present audit is the induced generalized-baker/Fourier program above.

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